

PROTECTION MECHANISMS For BATTERIES And Their CHARGING SYSTEMS

We take batteries for granted and often use them recklessly without taking care of them and their charging systems. This results in their shorter life and sometimes outright failure when we need them the most. The protection mechanisms described here could protect the batteries and their chargers even when these are misused



VISHAL SAPRE is an Electronics Design professional working with Righill Electronics as GM (Technical). His interests include embedded control systems, IoT, power electronics, and the use of interpreted language systems, such as Python in microcontrollers

Our modern-day lives are quite dependent on batteries, which come in all forms, shapes, and internal chemistries and are found almost everywhere. Batteries that can only convert their chemical energy into electrical energy and their electrochemical states cannot be revived are called primary batteries. Those whose electrochemical states can be revived back, using electrical current, are called secondary batteries or rechargeable batteries.

The secondary batteries are quite interesting in their electrical properties. When these are being discharged they behave like an energy source, but during charging they act as electrical loads. Rechargeable batteries come in different chemistries. The most prolific ones are the various types of lithium-ion (Li-ion) batteries found in almost all portable devices, such as mobile phones, laptops,

tablets, toys, etc. We shall be looking at the various battery protection mechanisms used in electronic circuits for such rechargeable batteries.

The lead-acid batteries are either sealed or tubular. These are also used very often in domestic, industrial, and automotive settings and, as of now, form the backbone of electrical energy storage applications, such as domestic and industrial inverters, emergency lighting, and various automotive applications.

The other two most used chemistries for rechargeable batteries are the NiMH (nickel metal hydride) and NiCd (nickel cadmium) batteries. These were till recently used in still photography cameras, however their use in these areas has diminished and taken over by Li-ion batteries. These are still used quite widely in railway applications as battery backup for the coaches.

TABLE 1
MAJOR PROTECTION MECHANISMS FOR BATTERIES AND THEIR CHARGERS

Protection Mechanism	Final Goal
Charging over-voltage protection	Avoid impressing charging voltage higher than the recommended maximum voltage for the battery.
Charging over-current protection	Avoid current flows higher than the maximum charging rate into the battery.
Over-temperature protection	Avoid situation where the battery temperature goes above its recommended maximum value.
Battery over-voltage protection	Avoid reverse current flow into the charger from the battery, if the battery voltage is higher than the charging voltage.
Reverse-polarity protection	Ensure connecting the battery to its charger in reverse polarity does not harm the charger, and indicates the incorrect connection to the user.
Short-circuit protection	Avoid prohibitively large current flow to take place across charger terminals. (This circuit shall be a part of the charger.) A separate battery management system is required to provide this protection at the battery terminals during discharging.
Deep-discharge protection	Avoid depleting the battery below a recommended minimum state of charge, or more than a given depth of discharge.